A8 - Gearbox Design

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Total time spent: **5 hours**.

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Objective

For this assignment, I worked from the required gear ratio and output conditions to size the shafts and journal bearings, then built a gearbox layout in SOLIDWORKS. I used the calculations to establish the shaft diameters and bearing requirements before arranging the gears inside a rectangular housing.

The gearbox uses an 18-tooth input pinion and a 48-tooth output gear. The larger gear increases torque while reducing speed. My final assembly view shows the gears and shafts inside the open housing so their arrangement is visible.

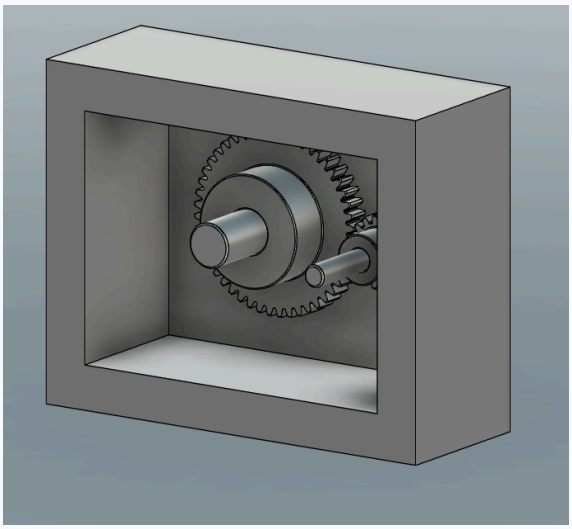


Figure 1. *Open housing assembly showing the 48-tooth gear, 18-tooth pinion, and protruding shafts. Full size.*

Design inputs and assumptions

Parameter	Input / pinion shaft	Output / gear shaft
Number of gear teeth, z	18	48
Pitch diameter, D_p	0.375 in	1.000 in
Shaft diameter, d	0.125 in	0.250 in
Shaft length, L	1.000 in	1.000 in
Speed, N	Calculated	1,100 rpm
Torque, T	Calculated	1.250 in·lbf
Allowable shear stress used	4,000 psi	8,500 psi

Both gears have a 20° pressure angle. The tooth ratio is 48:18, which simplifies to the required **24:9**. I treated the gear train as ideal, so transmitted power is the same at the input and output. Following the assignment, I ignored shaft bending from the gear forces for the shaft strength calculation and checked torsional shear stress.

For the bearing calculations, I used two supports per shaft and placed the gear load midway between them. This gives equal reactions at the two bearings. It is an analytical support model; the bearing reactions depend on the actual support and gear positions in a finished assembly.

The gear-to-shaft connection is intended to be a press fit. The shaft-to-bearing interface requires a running/sliding fit. These interfaces have different purposes: the gear must transmit torque with the shaft, while the bearing must allow the shaft to rotate.

Speed, torque, and power

The input speed follows from the gear ratio. With an ideal train, torque changes inversely with speed.

> $N_i z_i = N_o z_o$
> $N_i = 1{,}100(48/18) =$ **2,933.3 rpm**
>
> $T_i N_i = T_o N_o$
> $T_i = 1.250(18/48) =$ **0.46875 in·lbf**
>
> $H = TN / 63{,}025$
> $H_i = (0.46875)(2{,}933.3) / 63{,}025 =$ **0.02182 hp**
> $H_o = (1.250)(1{,}100) / 63{,}025 =$ **0.02182 hp**

The power check agrees at both ends. The input rotates faster and carries less torque than the output.

Input shaft design

Knowns, unknowns, and free-body diagram

The input shaft is 1.000 in long and 0.125 in in diameter. Its transmitted torque is 0.46875 in·lbf, and I used 4,000 psi as the allowable torsional shear stress. The unknowns are maximum shear stress and minimum required diameter.

For steady rotation, the shaft has equal and opposite end torques. Its torque balance is $T_{drive} - T_{gear} = 0$. The internal torque magnitude for this calculation is 0.46875 in·lbf. The shaft and support free-body diagrams are included in the handwritten work below.

Equations and result

For a solid circular shaft, c = d/2 and J = πd⁴/32. Substituting into τ = Tc/J gives:

> $\tau_{max} = 16T / (\pi d^3)$
> $\tau_{allow} \geq 16T / (\pi d^3)$
> $d_{min} = [16T / (\pi \tau_{allow})]^{1/3}$
>
> $\tau_i = 16(0.46875) / [\pi(0.125)^3] =$ **1,222.3 psi**
> $d_{min,i} = [16(0.46875) / (\pi \times 4{,}000)]^{1/3} =$ **0.0842 in**

The 0.125-in input shaft exceeds the calculated minimum diameter and its shear stress is below 4,000 psi. **I retained the specified 0.125-in diameter.** This conclusion applies to the torsion model and allowable stress used here.

Output shaft design

Knowns, unknowns, and free-body diagram

The output shaft is 1.000 in long and 0.250 in in diameter, with 1.250 in·lbf of torque at 1,100 rpm. I used 8,500 psi as its allowable shear stress. The unknowns are maximum shear stress and minimum diameter. The output shaft FBD has equal and opposite 1.250-in·lbf torques at its ends.

Equations and result

> $T_{gear} - T_{load} = 0$
> $\tau_o = 16(1.250) / [\pi(0.250)^3] =$ **407.4 psi**
> $d_{min,o} = [16(1.250) / (\pi \times 8{,}500)]^{1/3} =$ **0.0908 in**

The output diameter is greater than the calculated minimum, and the computed shear stress is below 8,500 psi. **I retained the specified 0.250-in diameter.**

Gear forces and bearing reactions

Force model

The known output torque is 1.250 in·lbf, the output pitch diameter is 1.000 in, and the pressure angle is 20°. I solved for tangential force F_t, radial force F_r, resultant gear load W, and bearing reactions R_A and R_B.

> $T_o = F_t(D_{p,o}/2)$
> $F_t = 2T_o/D_{p,o} = 2(1.250)/1.000 =$ **2.500 lbf**
> $F_r = F_t \tan \varphi = 2.500 \tan 20° =$ **0.9099 lbf**
> $W = \sqrt{(F_t^2 + F_r^2)} =$ **2.6604 lbf**

The meshing gears exert equal and opposite forces on each other. I used the magnitude of the resultant load for each shaft's bearing sizing calculation.

Support equilibrium

For a centered load over bearing span S, the moment equation about support A gives $R_B S = W(S/2)$. Force equilibrium then gives the other reaction.

> $\Sigma M_A = 0$: $R_B S - W(S/2) = 0$
> $R_B = W/2$
> $\Sigma F = 0$: $R_A + R_B - W = 0$
> $R_A = R_B = W/2 =$ **1.3302 lbf**
> $W_b =$ **1.3302 lbf per bearing**

The isolated bearing FBD balances the shaft load with the reaction from the housing. The load directions reverse between the input and output shaft diagrams, but the reaction magnitudes are equal in this centered-load model.

Journal bearing design

Material and reference limits

I used oil-impregnated sintered bronze as the bearing material for the calculations. The reference limits used in my work are shown below. These are material reference values, rather than a claim that every bronze sleeve bearing has the same rating. PIC Design material reference.

Property	Value used
Allowable projected pressure, P_{allow}	2,000 psi
Allowable sliding velocity, V_{allow}	1,200 ft/min
Allowable PV	50,000 psi·ft/min
Ultimate tensile strength, S_{ut}	14,000 psi
Ultimate-strength design factor, n_u	4
Allowable hoop stress, S_{ut}/n_u	3,500 psi

Length, pressure, velocity, and PV

The unknowns are the minimum bearing length, operating pressure, sliding velocity, and PV. I used projected area dL for the pressure calculation. The minimum length must satisfy both the pressure and PV limits; the velocity limit is checked separately.

> $A_{projected} = dL$
> $P = W_b/(dL)$
> $V = \pi dN/12$ [d in inches; V in ft/min]
> $PV = W_bV/(dL)$
>
> $L_P \geq W_b/(dP_{allow})$
> $L_{PV} \geq W_bV/[d(PV)_{allow}]$
> $L \geq \max(L_P, L_{PV})$

For the input bearing:

> $V_i = \pi(0.125)(2{,}933.3)/12 =$ **95.99 ft/min**
> $L_{P,i} \geq 1.3302/[(0.125)(2{,}000)] =$ **0.00532 in**
> $L_{PV,i} \geq (1.3302)(95.99)/[(0.125)(50{,}000)] =$ **0.02043 in**

For the output bearing:

> $V_o = \pi(0.250)(1{,}100)/12 =$ **71.99 ft/min**
> $L_{P,o} \geq 1.3302/[(0.250)(2{,}000)] =$ **0.00266 in**
> $L_{PV,o} \geq (1.3302)(71.99)/[(0.250)(50{,}000)] =$ **0.00766 in**

I used **0.250 in as the nominal bearing length** for both shaft sizes. At that length, the operating values are:

Check	Input bearing	Output bearing	Reference limit
Journal diameter	0.125 in	0.250 in	Shaft size
Nominal length	0.250 in	0.250 in	Above calculated minima
Projected pressure	42.57 psi	21.28 psi	2,000 psi
Sliding velocity	95.99 ft/min	71.99 ft/min	1,200 ft/min
PV	4,086 psi·ft/min	1,532 psi·ft/min	50,000 psi·ft/min

All three operating checks are below the reference limits. The input bearing controls the length calculation because its smaller journal and higher speed produce the larger PV requirement.

Bearing wall thickness

I modeled the sleeve as a thick cylinder with a uniform equivalent internal pressure p and zero external pressure. I used the projected bearing pressures above for p. This is a simplified wall-stress calculation; it does not include the actual contact-pressure distribution, interference with the housing, or thermal effects.

The knowns are journal diameter D_i, pressure p, and allowable hoop stress σ_{allow} = 3,500 psi. The unknowns are minimum outside diameter $D_{o,min}$ and minimum radial wall thickness t_{min}.

> $\sigma_{\theta,max} = p(D_o^2 + D_i^2)/(D_o^2 - D_i^2)$
> $(\sigma_{allow} - p)D_o^2 = (\sigma_{allow} + p)D_i^2$
> $D_{o,min} = D_i\sqrt{[(\sigma_{allow} + p)/(\sigma_{allow} - p)]}$
> $t_{min} = (D_{o,min} - D_i)/2$
>
> $t_{min,i} = (0.125/2)\{\sqrt{[(3{,}500 + 42.57)/(3{,}500 - 42.57)]} - 1\}$
> $t_{min,i}$ = **0.000765 in**
>
> $t_{min,o} = (0.250/2)\{\sqrt{[(3{,}500 + 21.28)/(3{,}500 - 21.28)]} - 1\}$
> $t_{min,o}$ = **0.000762 in**

These small values are theoretical stress minima. They are not practical wall thicknesses to manufacture. A catalog sleeve needs enough material for handling, installation, and its specified housing fit as well as the calculated load.

For component selection, the bearing types are **oil-embedded bronze sleeve bearings for a 1/8-in shaft** and **oil-embedded bronze sleeve bearings for a 1/4-in shaft**, with a nominal 1/4-in length for this calculation. The supplier's outside diameter and installed-bore requirements determine the corresponding housing seat. McMaster-Carr sleeve bearings.

Housing sketch and assembly concept

My sketch uses a rectangular housing body with a removable front component. The two attachment locations shown above and below the opening establish the intended screw-fastened connection. A removable front gives access to the gears and shafts during assembly and disassembly.

I considered making the housing follow the gear outlines more closely, but kept the rectangular layout for this model. It gives a simple starting shape for the cavity and leaves a rim around the front opening. The sketch records the enclosure concept; the open CAD view below shows the internal arrangement.

Parametric CAD modeling

Shafts and gear placement

I modeled the shafts as dimensioned circular profiles extruded to 1.000 in. The input diameter is 0.125 in and the output diameter is 0.250 in. The shaft drawings include 0.010-in × 45° end chamfers. The input screenshot rounds the displayed diameter to 0.13 in; the detailed drawing records the intended 0.125-in dimension.

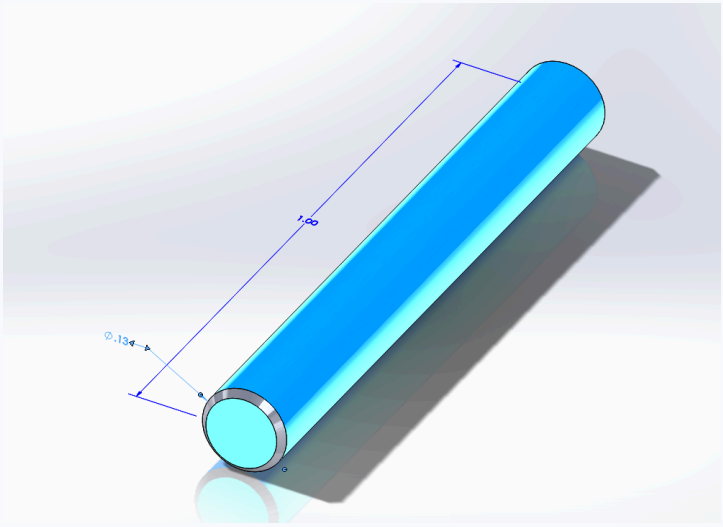


Figure 9. *Input shaft CAD: 1.000-in length. The diameter display is rounded; the drawing specifies 0.125 in. Full size.*

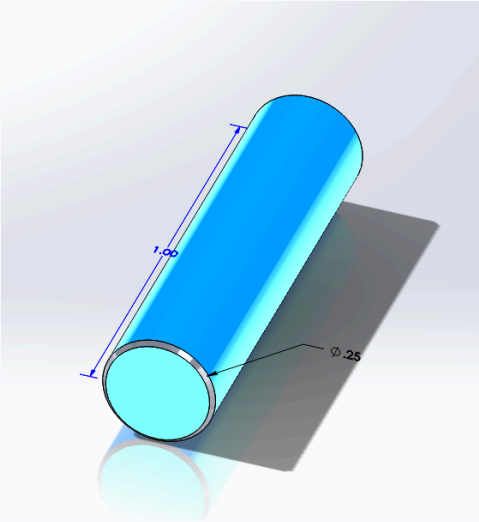


Figure 10. *Output shaft CAD: 1.000-in length and 0.250-in diameter, with chamfered ends. Full size.*

I used the McMaster-Carr models for the 18-tooth pinion, **7880K19**, and 48-tooth gear, **7880K24**. Their pitch diameters establish the shaft center distance:

> $C = (D_{p,i} + D_{p,o})/2$
> $C = (0.375 + 1.000)/2 =$ **0.6875 in**

I placed the shaft axes 0.6875 in apart. Using the outside diameters for this distance would space the gears incorrectly because the pitch circles establish the nominal mesh. The screenshot records the center-distance dimension used in the layout.

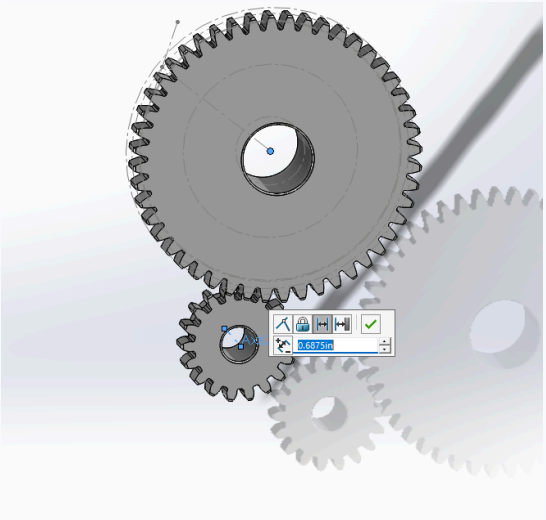


Figure 11. *Gear layout with the 0.6875-in center-distance dimension. Full size.*

Housing dimensions and design intent

The housing drawing records a **2.400-in width, 1.900-in height, and 0.900-in overall depth**. The front opening is surrounded by a 0.250-in rim, and the back-wall callout is 0.100 in. I used the gear arrangement to establish the cavity position and placed the shaft openings in the rear wall.

The important relationships are the fixed shaft center distance, the shaft diameters, the cavity around the gear teeth and hubs, and the bearing seats. A clearance around the tooth tips is separate from the fit at the rotating journals. The final view communicates the component arrangement; it is not a physical rotation or interference test.

CAD drawings

The drawings document the two shafts and the housing body. Each preview opens the original PDF so the dimensions can be read at full size. They are also reproduced in the submission report's drawing appendix.

Results

Design quantity	Input side	Output side
Speed	2,933.3 rpm	1,100 rpm
Torque	0.46875 in·lbf	1.250 in·lbf
Transmitted power	0.02182 hp	0.02182 hp
Calculated torsional shear stress	1,222.3 psi	407.4 psi
Minimum shaft diameter	0.0842 in	0.0908 in
Retained shaft diameter	0.125 in	0.250 in
Load per bearing, centered-support model	1.3302 lbf	1.3302 lbf
Minimum length from pressure and PV	0.02043 in	0.00766 in
Nominal length used	0.250 in	0.250 in
Theoretical minimum sleeve wall	0.000765 in	0.000762 in

Both shafts meet the torsional strength checks with the specified diameters. A 0.250-in bearing length satisfies the pressure, speed, and PV checks for the bronze reference data and assumed bearing loads. The CAD layout uses the calculated 0.6875-in gear center distance and shows how the shafts and gears fit within the open rectangular housing.

Lessons learned and mistakes

Pitch diameter controls gear spacing. I had to distinguish the pitch circles from the outside tooth diameters when positioning the gears. Adding the pitch radii gave a clear center-distance dimension to use in CAD.

A diameter display can hide useful precision. The input-shaft screenshot shows 0.13 in after rounding, while the intended dimension is 0.125 in. I used the detailed drawing to communicate the shaft size rather than relying on the rounded screenshot alone.

Strength and fit answer different questions. A shaft can pass a torsion calculation while still requiring a properly specified fit at each interface. The gear hub needs to transmit torque, and the bearing bore needs clearance for rotation. Treating every hole as the same nominal shaft diameter would miss that distinction.

Bearing pressure uses projected area. I used dL rather than the full curved surface area. The input bearing also showed why pressure alone is insufficient: its higher speed makes the PV check more restrictive than the pressure-only length check.

A theoretical minimum is not a manufacturing dimension. The wall-stress calculation produced thicknesses below one thousandth of an inch because the load is small relative to the assumed material strength. A usable sleeve still needs a practical wall and an appropriate installed fit.

Assembly details need their own dimensions. Working through the housing made the relationship between shaft position, gear hubs, bearing support, and the removable front more apparent. The equations establish the operating loads, while the component layout determines where those loads enter the housing.

Total time spent on calculations, sketching, CAD, and documentation: **5 hours**.

CAD and source downloads

File	Contents
Assembly export	Final exported mesh geometry for viewing the assembly
CAD source package	Two shaft parts, two gear parts, gear STEP files, the working assembly saves, and the 3MF export
Input shaft drawing	Original 0.125-in shaft drawing
Output shaft drawing	Original 0.250-in shaft drawing
Housing body drawing	Original housing drawing
Original work and screenshots	All seven handwritten pages and four CAD screenshots
Submission report	Report with full-page handwritten work and original drawing appendices

The 3MF is a mesh export. The source package retains the original native SOLIDWORKS component and working assembly filenames so their references can be kept together.

References

- Course assignment: *A8: Gear Box Design*, MEGR 2157. Required output conditions, shaft dimensions, assumptions, and drawing requirements.
- *Machinery's Handbook*: shaft torsion, gearing, and plain-bearing methods referenced by the assignment.
- McMaster-Carr 7880K19 pinion and 7880K24 gear: gear component models and nominal geometry.
- PIC Design bronze bearing reference: material data used for the bearing pressure, velocity, PV, and strength assumptions.
- McMaster-Carr journal sleeve bearings: sleeve-bearing component selection.

Appendix A - Handwritten work 1 of 7

Handwritten input-speed, torque, power, and shaft-design setup.

Input Speed, Torque, & Transmitted power

Knowns:

$d_i = 0.125\,in$ $\quad N_i = 1100\,rpm$ $\quad d_o = 0.250\,in$

$L_i = 1.00\,in$ $\quad \tau_{allow,i} = 4000\,psi$

$T_i = 0.46875\,in\,lbf$ $\quad z_i = 18$ $\quad z_o = 48$ $\quad i = \frac{z_o}{z_i} = \frac{48}{18} = \frac{24}{9}$

$D_{p,i} = 0.375\,in$ $\quad D_{p,o} = 1.00\,in$

$\phi = 20°$

Unknowns:

N_i, T_i, H

Input speed:

$N_i z_i = N_o z_o$

$N_i = N_o \frac{z_o}{z_i}$

$N_i = 1100\left(\frac{48}{18}\right)$

$N_i = 2933.3\,rpm$

Input Torque

$T_i N_i = T_o N_o$

$T_i = T_o \frac{N_o}{N_i}$

$T_i = 1.25\left(\frac{1100}{2933.3}\right)$

$T_i = 0.46875\,in\cdot lbf$

Transmitted power:

$H = \frac{TN}{63025}$

$H_i = \frac{(0.46875)(2933.3)}{63025}$

$H_i = 0.02182\,hp$

$H_o = \frac{(1.25)(1100)}{63025}$

$H_o = 0.02182\,hp$

$H_i = H_o$

Input - Shaft strength

Knowns:

$T_i = 0.46875\,in{\cdot}lbf$

$d_i = 0.125\,in$ $\quad L_i = 1.000\,in$

$N_i = 2933.3\,rpm$

$\tau_{allow,i} = 4000\,psi$

Unkowns:

τ_i , $d_{min,i}$

Appendix A - Handwritten work 2 of 7

Handwritten torsion equations, shaft FBD, and gear-force calculations. The typed shaft sections separate the input and output results.

FBD:



$L = 1.00\,in$ $\quad d = 0.125\,in$ $\quad T_1 = 0.46875\,in\cdot lbf$ $\quad T_2 = 0.4$

$\sum T = 0$

$T_{gear} - T_{load} = 0$

$$\tau_{max} = \frac{Tc}{J}, \quad \tau_{max} = \frac{T\left(\frac{d}{2}\right)}{\pi\frac{d^4}{32}} = \frac{16T}{\pi d^3}$$

$$\tau_o = \frac{16(1.25)}{\pi(0.250)^3} = 407.4\,psi$$

$$\tau_{allow,o} = \frac{16T_o}{\pi d_{min,o}^3} \qquad d_{min,o} = \left(\frac{16T_o}{\pi\tau_{allow,o}}\right) \qquad d_{min,o} = \left(\frac{16(1.25)}{\pi(8500)}\right)^{1/3} \qquad d_{min,o} = 0.0908\,in$$

Gear Forces & bearing reactions:

Knowns:

$T_o = 1.25\,in\cdot lbf$ $\quad \phi = 20^\circ$

$D_{p,o} = 1.000\,in$

Unknowns:

F_t, F_r, W, R_A, R_B

$$r_{p,o} = \frac{D_{p,o}}{2} = 0.500\,in$$

$$T_o = F_t r_{p,o} \qquad F_t = \frac{T_o}{r_{p,o}} = \frac{2T_o}{D_{p,o}}$$

$$F_t = \frac{2(1.25)}{1.000} \qquad F_t = 2.500\,lbf$$

$$F_r = F_t \tan\phi$$

$$F_r = 2.500\tan 20^\circ$$

$$F_r = 0.9099\,lbf$$

$$W = \sqrt{F_t^2 + F_r^2}$$

$$W = \sqrt{2.5^2 + 0.9099^2}$$

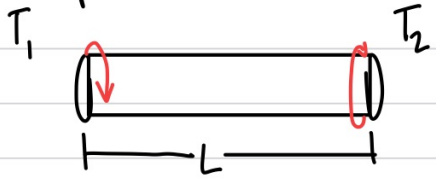
$$W = 2.6604\,lbf$$

Mechatronics Homework-2.jpg

Appendix A - Handwritten work 3 of 7

Output shaft torque FBD, centered bearing-support diagrams, and individual bearing equilibrium.

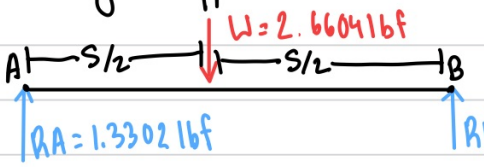
Output Shaft FBD



$T_1 = 1.250\ in \cdot lbf$
$T_2 = 1.250\ in \cdot lbf$
$d = 0.250\ in$
$L = 1.00\ in$

Bearing support FBD

Input Shaft

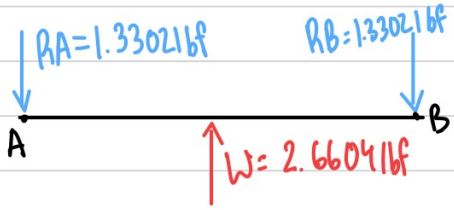


$\sum M_A = 0$

$R_B S - W\frac{S}{2} = 0$

$R_B = \frac{W}{2}$

Output Shaft



$\sum F = 0$

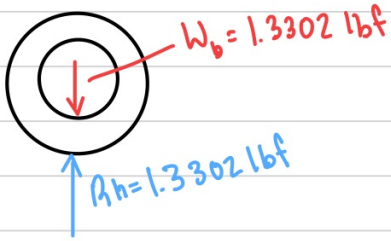
$R_A + R_B - W = 0$

$R_A = W - R_B = \frac{W}{2}$

$R_A = R_B = \frac{2.6604}{2}$

$W_b = R_A = R_B = 1.3302\ lbf$

Individual Bearing FBD



$\sum F = 0$

$R_h - W_b = 0$

$R_h = W_b$

Appendix A - Handwritten work 4 of 7

Bronze reference limits and input bearing length from pressure and PV.

Bearing material & refrence limits

$P_{allow} = 2000\ psi$ $S_{ut} = 14000\ psi$

$V_{allow} = 1200\ ft/min$

$(PV)_{allow} = 50\,000\ \frac{psi.ft}{min}$ $n_u = 4$

$\sigma_{allow} = \frac{S_{ut}}{n_u}$ $\sigma_{allow} = \frac{14000}{4} = 3500\ psi$

Input journal bearing length

Knowns:

$W_b = 1.3302$ $N = 2933.3\ rpm$

$d = 0.125\ in$ $P_{allow} = 2000\ psi$

$V_{allow} = 1200\ \frac{ft}{min}$ $PV_{allow} = 50000\ \frac{psi.ft}{min}$

Unknowns:

L_{min}, P, V, PV

$A_{proj} = dL$ $P = \frac{W_b}{A_{proj}} = \frac{W_b}{dL}$

$V = \frac{\pi d N}{12}$ $V = \frac{\pi(0.125)(2933.3)}{12}$ $V = 95.99\ \frac{ft}{min}$

$P_{allow} \geq \frac{W_b}{dL}$ $L_p \geq \frac{W_b}{d P_{allow}}$ $L_p \geq \frac{1.3302}{(0.125)(2000)}$ $L_p \geq 0.00532\ in$

$PV = \frac{W_b V}{dL}$ $(PV)_{allow} \geq \frac{(1.3302)(95.99)}{(0.125)(50000)}$ $L_{pv} \geq 0.02043\ in$

Appendix A - Handwritten work 5 of 7

Output bearing length and setup for the sleeve wall-thickness model.

Output journal bearing Length

Knowns

$W_b = 1.3302\ lbf$ $N = 1100\ rpm$
$d = 0.250\ in$ $P_{allow} = 2000\ psi$
$V_{allow} = 1200\ ft/min$ $(PV)_{allow} = 50000\ \frac{psi\text{-}ft}{min}$

Unknowns:
L_{min}, P, V, PV

$$V = \frac{\pi d N}{12} \qquad V = \frac{\pi (0.250)(1100)}{12}$$
$$V = 71.99\ ft/min$$

$$L_p \geq \frac{W_b}{d P_{allow}} \qquad L_p \geq 0.00266\ in$$

$$L_{pv} \geq \frac{W_b V}{d (PV)_{allow}} \qquad L_{pv} \geq 0.0076\ in$$

$$P_o = \frac{W_b}{dL} \qquad P_o = 21.28\ psi$$

Bearing wall thickness model

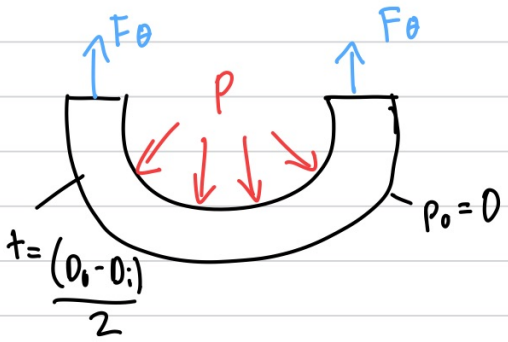
Knowns:
$D_{i,input} = 0.125\ in$ $p_{output} = 21.28\ psi$
$D_{i,output} = 0.250\ in$
$p_{input} = 42.57\ psi$ $\sigma_{allow} = 3500\ psi$

Unknowns:
$D_{o,min}$
t_{min}

Appendix A - Handwritten work 6 of 7

Sleeve wall-stress FBD, thick-cylinder model, and input/output thickness calculations.

FBD:



$$\sigma_{\theta,max} = p\left(\frac{D_o^2 + D_i^2}{D_o^2 - D_i^2}\right)$$

$$\sigma_{allow} D_o^2 - \sigma_{allow} D_i^2 = D_i^2(\sigma_{allow} + p)$$

$$D_{o,min} = D_i\sqrt{\frac{\sigma_{allow} + p}{\sigma_{allow} - p}}$$

$$t_{min} = \frac{D_i}{2}\left(\sqrt{\frac{\sigma_{allow} + p}{\sigma_{allow} - p}} - 1\right)$$

Input bearing Wall thickness

$D_i = 0.125\,in$

$p = 42.5671\,psi$ $\quad t_{min} = 0.000765\,in$

$\sigma_{allow} = 3500\,psi$

Output bearing Wall thickness

$D_i = 0.25\,in$ $\quad \sigma_{allow} = 3500\,psi$

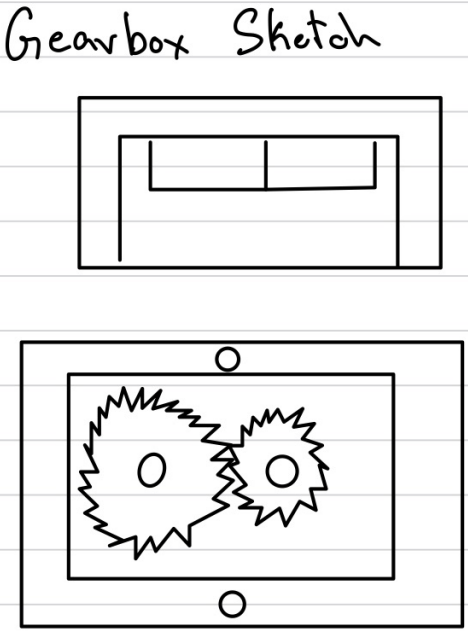
$p = 21.2836\,psi$

$t_{min,o} = 0.000762\,in$

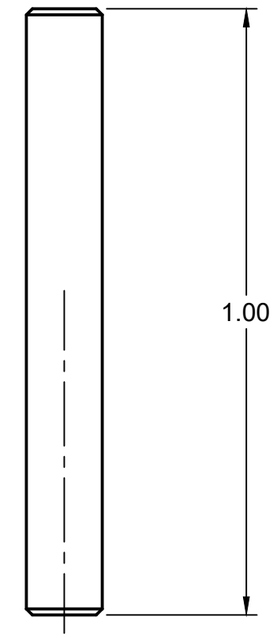
Mechatronics Homework-6.jpg

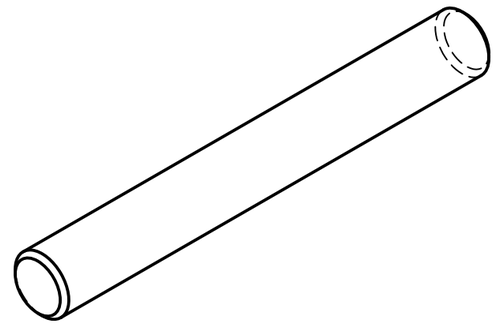
Appendix A - Handwritten work 7 of 7

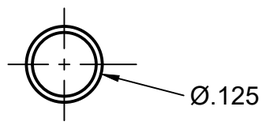
Housing sketch showing the gear pair, enclosure, and two front attachment locations.

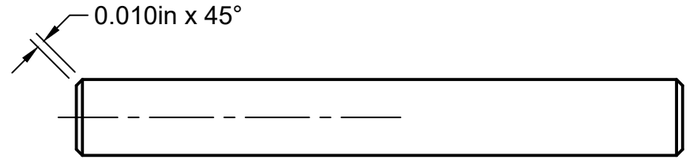


Mechatronics Homework-7.jpg

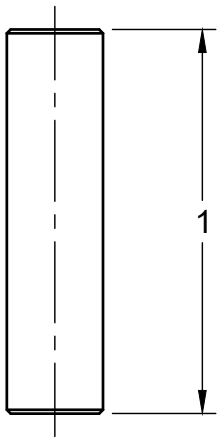


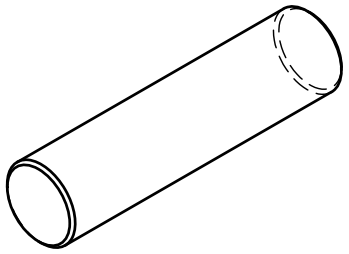


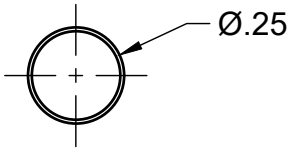


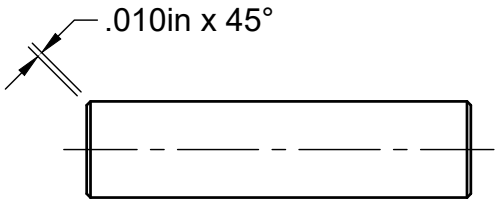


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APPROVED	SIZE B	CODE ASME Y14.5	DWG NO	REV
CHECKED				
DRAWN Rick Bhattacharya 10/29/2026	SCALE 4:1	WEIGHT	SHEET 1/1	

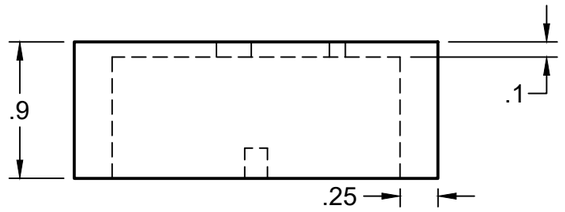


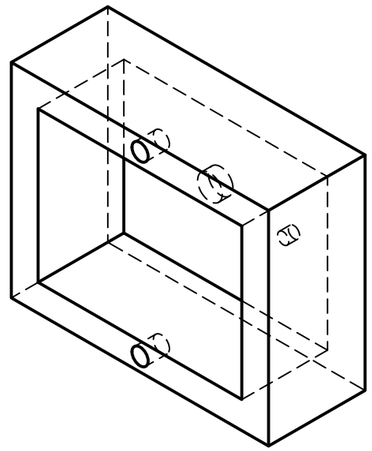


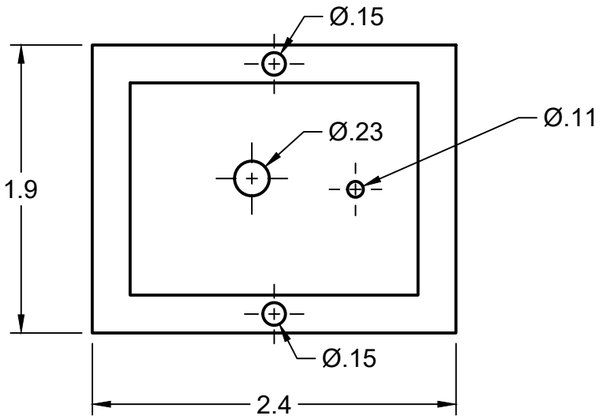


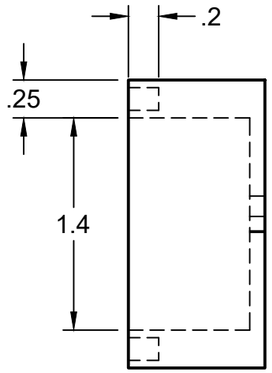


	PROJECT A8			
	TITLE SHAFT 2			
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	A	ASME Y14.5		
DRAWN Rick Bhattacharya 10/29/2026	SCALE 2:1	WEIGHT	SHEET 1/1	









	PROJECT A8			
	TITLE SHAFT 2			
APPROVED	SIZE B	CODE ASME Y14.5	DWG NO	REV
CHECKED				
DRAWN Rick Bhattacharya 10/29/2026	SCALE 1:1	WEIGHT	SHEET 1/1	